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1 A BENGALURU-BASED DEEP-TECH STARTUP KILLS GERMS IN MID-AIR

A deep-tech startup is addressing airborne infection risks through research-driven air decontamination technology.

Air in hospitals is meant to protect lives, yet it often becomes a silent carrier of infections. While most air purifiers focus on trapping dust and particles, a Bengaluru-based deep-tech startup has developed a device that destroys airborne microbes as they pass through it. Founded in 2014 by scientists Dr Janani Venkatraman, Dr Arindam Ghatak, and Dr Santanu Datta, Biomoneta Research Private Limited seeks to make clinically proven, science-backed air safety a standard of care in environments that protect vulnerable lives.



Biomoneta Avata Rx

The startup commercialises its innovation through products such as Avata Rx, embedding the ZeB-box platform into devices designed for specific, high-need environments rather than offering it as a standalone unit.

Dr Janani explains that the novelty of the ZeB-box platform lies in integrating three critical elements into a single, intelligent system. At the core are proprietary microbicidal surfaces that inactivate microorganisms upon contact.

This is supported by a carefully engineered airflow architecture that guides airborne microbes towards these active surfaces, preventing them from slipping past, as often happens in conventional systems. The third component is precision power electronics that generate a finely tuned electric field designed around the natural behaviour of microbes.

Since all pathogens—whether bacteria, viruses, or fungi—carry surface charges, ZeB-box leverages this universal trait by generating an electric field that pulls microbes out of

the air and into the microbicidal surface, where they are destroyed in real time. This approach differentiates ZeB-box from HEPA filters that merely trap particles, UV systems that struggle under real-world airflow conditions, and plasma- or ozone-based technologies that risk generating harmful by-products.

Unlike consumer-grade air purifiers, ZeB-box products are engineered to meet stringent safety and performance standards aligned

with their intended use cases.

Devices designed for healthcare environments comply with IEC 60601 medical device standards, while models for consumer spaces adhere to IEC 60335 electrical safety requirements. As a result, the solutions are deployed across ICUs, NICUs, oncology wards, operation theatres, OPDs, dialysis centres, IVF labs, and dental clinics, where controlling airborne infection is critical.

Beyond healthcare, Biomoneta serves pharmaceutical and biotech cleanrooms that require strict microbial control to meet regulatory standards, as well as high-risk home-care settings for immunocompromised patients. The platform also supports specialised integrations for ambulances, HVAC and AHU systems, office cubicles, and high-end automobiles, highlighting its adaptability across both compact and large-scale applications.

The startup manufactures its products in-house at an ISO 13485-certified facility in Sahakar Nagar, Bengaluru. Its compact manufacturing team of around ten engineers and technicians is supported by basic tooling, 3D printing, and automated assembly equipment.

The ZeB-box is a patented 'trap and kill' air decontamination platform that actively attracts and destroys airborne microbes, including bacteria, fungi, viruses, and spores. Designed as a flexible platform, it enables the development of multiple devices tailored to environments ranging from hospital